

# Prototyping and Proof of Concept for AI Projects

Instructor: Dr Farnaz Derakhshan  
Seneca Polytechnic

# Session Overview

- Prototyping
- Data Collection
- Proof of Concept (POC)
- Pipeline thinking

# What is Prototyping

- Early model/system version
- Validates idea quickly
- Low cost experimentation

# Why Prototyping Matters

- Reduces risk
- Early feedback
- Faster iteration

# Types of Prototypes



MODEL  
PROTOTYPE



SYSTEM  
PROTOTYPE



DATA PROTOTYPE

## Discussion

- Do you have any experience in Prototyping? Share your experiences

# AI Development Flow



DATA → MODEL →  
DEPLOYMENT



ITERATIVE  
PROCESS



VALIDATION  
CYCLES

# Data Collection Basics

- Define problem
- Identify data sources
- Ensure relevance

# Data Sources

- Public datasets
- APIs
- Internal data
- Simulation

# Sampling Data

- Use subset of dataset
- Faster experiments
- Avoid full dataset initially

# Exploratory Data Analysis

- Understand distributions
- Visualize data
- Detect anomalies

Note: Consider data quality regarding missing values, Noise, Bias, and Consistency

# Importing Data

- CSV files
- JSON
- Databases
- APIs

# Tools for Importing

- Pandas
- SQL
- Cloud storage
- ETL tools

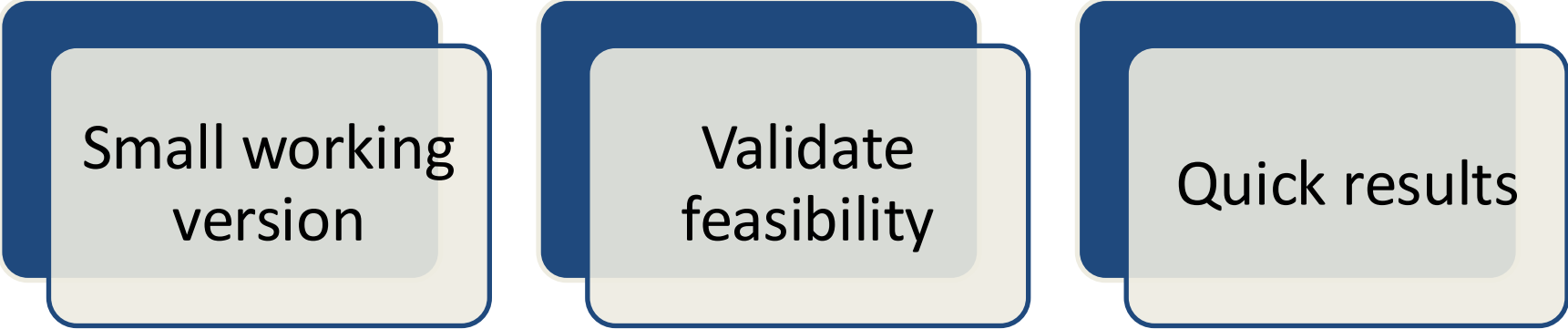
# Data Preprocessing

- Cleaning
- Normalization
- Feature engineering
  - Transform data
  - Encode variables
  - Create features

# Train/Test Split

- Separate data
- Avoid overfitting
- Evaluate model

# Proof of Concept (POC)

The diagram consists of three identical, horizontally aligned rectangular boxes. Each box is composed of three overlapping layers: a dark blue top layer, a light gray middle layer, and a light beige bottom layer. The boxes are slightly offset to the right, creating a sense of depth. The text is centered within the light gray middle layer of each box.

Small working  
version

Validate  
feasibility

Quick results

# Why POC Important

- Test assumptions
- Reduce uncertainty
- Guide project

# POC vs Final System

- Simple vs scalable
- Fast vs optimized
- Experimental

# Building a POC

1. Select small dataset
2. Use simple model
3. Evaluate quickly

# Model Selection (POC)

- Start simple
- Use baseline:

A baseline is a simple model or approach used as a reference point to compare more complex models.

- Compare results

# Evaluation Metrics

- Accuracy
- Precision/Recall
- RMSE
- F1 Score

# Visualization



GRAPHS



CHARTS



INSIGHTS



STORYTELLING

# Iteration Process



Test → Evaluate → Improve



Continuous learning

# Common Challenges



DATA ISSUES



MODEL  
PERFORMANCE



TIME  
CONSTRAINTS

# Best Practices



KEEP IT SIMPLE



DOCUMENT STEPS



VALIDATE  
CONTINUOUSLY

# Case Study Example: Fraud Detection POC in Banking

## **1. Problem (Applied Gap)** A bank wants to:

- Detect fraudulent transactions
- Reduce financial loss
- Improve customer trust

Challenge:

- Huge number of transactions
- Very few fraud cases (imbalanced data)

## **2. Goal of the POC**

Test whether machine learning can detect fraud effectively using a small sample of data.

## **3. Step 1: Data Collection** (Sample Only)

Instead of full dataset (millions of records):

Select 10,000 transactions

Include: transaction amount, time, location, and fraud label

Reason: Faster experimentation, and Lower computational cost

# Case Study Example: Fraud Detection POC in Banking

## 4. Step 2: Data Preparation

- Clean and prepare data:
- Handle missing values
- Normalize amount
- Encode categorical
- **featuresExample:**
- **Plain TextLocation → One-hot encodingAmount → Normalized**

## 5. Step 3: Exploratory Analysis

- Students should check:
- Fraud percentage (usually <1%)
- Distribution of transaction amounts
- Patterns (large transactions at unusual times)

## 6. Step 4: Build Baseline Model

Start simple: Logistic Regression or Decision Tree

**This is the baseline**

# Case Study Example: Fraud Detection POC in Banking

**7. Step 5:** Build POC Model, Try improved model: Random Forest or XGBoost

**8. Step 6:** Evaluation

- Use metrics like: Accuracy, Recall, F1 Score
- Important for fraud: High recall (detect fraud cases)
- **Example**

| Model     | Accuracy | Recall (Fraud) |
|-----------|----------|----------------|
| Baseline  | 85%      | 40%            |
| POC Model | 92%      | 75%            |

- Insight: Model significantly improves fraud detection
- **9. Step 7: POC Outcome**, the team decides: ✓ ML approach is feasible ✓ More data may improve results ✓ Ready to move to next phase.
- **10. Limitations of POC Small dataset** Not production-ready No real-time deployment

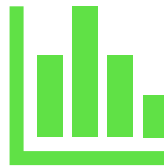
## Class Activity

- Build mini POC for your Capstone Project.
- Use sample dataset
- Discuss outcomes

# Key Takeaways



Prototype early



Use sample data



Build POC first